

DEPI Assignment 1

Task 1

Calculate and display a grade summary for a student:

1. Create list of grades using appropriate data type
2. Calculates the total grades using for-in loop
3. Calculates average
4. Determines performance level based on the average using switch expression:
 - a. Excellent for 90-100
 - b. Good for 80-90
 - c. Average for 70-80
 - d. Bad for grades below 70
5. Prints a formatted summary using only one print.

Output Sample:

```
Grade Summary:  
Total Grades: 4  
Average: 86.00  
Performance: Good
```

Task 2

Q1

What's the output?

dart

```
void main() {  
  String name = 'Ahmed';  
  int age = 25;  
  double height = 1.75;  
  bool isStudent = true;  
  
  print('$name is $age years old, $height meters tall. Student: $isStudent');  
}
```

- A) Ahmed is 25 years old, 1.75 meters tall. Student: true
- B) name is age years old, height meters tall. Student: isStudent
- C) Ahmed is 25 years old, 1.75 meters tall. Student: 1
- D) Ahmed is 25 years old, 1.75 meters tall. Student: True

Q2

What's the output?

dart

```
void main() {  
  var x = 10;  
  dynamic y = 'Hello';  
  y = 20;  
  print('${x.runtimeType} ${y.runtimeType}');  
}
```

- A) int String
- B) int int
- C) var dynamic
- D) 10 20

Q3

dart

```
void main() {  
  const PI = 3.14;  
  final currentTime = DateTime.now();  
  print(PI);  
  print(currentTime);  
  PI = 3.14159;  
  currentTime = DateTime.now();  
}
```

- A) Line 6 causes error
- B) Line 7 causes error
- ☒ C) Both lines cause errors
- D) Neither line causes error

Q4

dart

```
void main() {  
  String? firstName;  
  String? lastName = 'Smith';  
  int? age;  
  
  String fullName = (firstName ?? 'John') + ' ' + (lastName ?? 'Doe');  
  int displayAge = age ?? 18;  
  
  print('$fullName, $displayAge years old');  
}
```

- ☒ A) John Smith, 18 years old
- B) null Smith, null years old
- C) John Doe, 18 years old
- D) Smith, 18 years old

Q5

```
dart

void main() {
  const List<int> constNumbers = [1, 2, 3];
  final List<int> finalNumbers = [1, 2, 3];

  constNumbers.add(4);
  finalNumbers.add(4);

  print(finalNumbers);
}
```

A) Line 5 causes error
B) Line 6 causes error
C) Both lines cause errors
D) Neither line causes error

Q6

```
dart

void main() {
  String day = 'Wednesday';
  String dayType;

  switch (day) {
    case 'Saturday':
    case 'Sunday':
      dayType = 'Weekend';
      break;
    case 'Monday':
    case 'Tuesday':
    case 'Wednesday':
    case 'Thursday':
    case 'Friday':
      dayType = 'Weekday';
      break;
    default:
      dayType = 'Unknown';
  }

  print('$day is a $dayType');
}
```

A) Wednesday is a Weekday
B) Wednesday is a Weekend
C) Wednesday is a Unknown
D) Error: missing break statement

Q7

```
dart

void main() {
  int score = 85;
  String grade = switch(score) {
    >= 90 => 'A',
    >= 80 => 'B',
    >= 70 => 'C',
    >= 60 => 'D',
    _ => 'F',
  };

  print('Score: $score, Grade: $grade');
}
```

A) Score: 85, Grade: B
B) Score: 85, Grade: A
C) Score: 85, Grade: C
D) Error: invalid syntax

Q8

```
dart

void main() {
  bool hasMembership = true;
  double price = 100.0;

  double finalPrice = hasMembership ? price * 0.8 : price;
  String discountText = hasMembership ? '20% discount' : 'No discount';

  print('Price: \${finalPrice.toStringAsFixed(2)} ($discountText)');
}
```

A) Price: \$80.00 (20% discount)
B) Price: \$100.00 (No discount)
C) Price: \$80.00 (No discount)
D) Price: \$100.00 (20% discount)

Q9

dart

```
void main() {  
  String? text;
```

```
  print(text.length);
```

```
  print(text?.length);
```

```
  print(text!.length);  
}
```

- A) Line 4 causes error, line 5 prints null, line 6 causes error
- B) Line 4 prints 0, line 5 prints 0, line 6 prints 0
- C) Line 4 prints null, line 5 prints null, line 6 prints null
- D) All lines work correctly

Q10

dart

```
void main() {
```

```
  var mixedList = [1, 'two', 3.0, true];
```

```
  print('${mixedList[0]} is ${mixedList[0].runtimeType}');
```

```
  print('${mixedList[3]} is ${mixedList[3].runtimeType}');
```

```
}
```

- A) 1 is int true is bool
- B) 1 is dynamic true is dynamic
- C) 1 is var true is var
- D) Error: mixed types not allowed

Q11

What's the output?

dart

```
const int maxAttempts = 3;
var attempts = 0;
String status = '';

do {
  attempts++;
  status = switch(attempts) {
    1 => 'First try',
    2 => 'Second try',
    _ => 'Last try',
  };
  print(status);
} while (attempts < maxAttempts);

print('Total: $attempts');
```

- A) First try Second try Total: 2
-  B) First try Second try Last try Total: 3
- C) First try Second try Third try Total: 3
- D) Infinite loop